EmbMerge: A Transformer-based Method for Fusing CDR Lists

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Abstract. To overcome the problem of data sparsity in recommendation systems, cross-domain recommendation (CDR) models leverage user data from a source domain to improve recommendations in a target domain. However, different CDR models capture unique and often complementary aspects of user preferences. This paper introduces EmbMerge, a supervised method to fuse the outputs of these diverse CDR models to produce a better, unified ranking. Traditional fusion methods are mainly based on sparse features that do not capture deep semantic relationships. EmbMerge instead employs rather small dense vector representations for models, items, ranks, and scores, processed by a transformer encoder to generate rich, context-aware embeddings. We propose two architectural variants and conduct experiments on two datasets using ranked lists from three different state-of-the-art CDR systems. Our results demonstrate that EmbMerge outperforms four baseline fusion methods, validating its effectiveness as a technique for combining the strengths of various cross-domain recommendation systems.

Keywords: Cross-Domain Recommendation · Rank Fusion · Learning to Rank

1 Introduction

Modern recommendation systems often suffer from data sparsity and cold-start problems, where it is difficult to make accurate recommendations for new users or items with limited interaction data. Cross-Domain Recommendation (CDR) methods [9,18,22,14,16,20,8] have emerged as a solution to this challenge. The core idea of CDR is to transfer knowledge about a user's preferences from a source domain (e.g., book ratings) to improve recommendations in a data-sparse target domain (e.g., movie suggestions). Using this information, CDR systems can significantly decrease data sparsity and improve the quality of recommendations.

However, in the field of CDR various models exist, each with different architectures. Consequently, these different CDR models often capture distinct signals

about user preferences. One model might succeed in identifying item-level feature similarities across domains, while another might be better at modeling a user's general taste. No single CDR model can perfectly capture the complete user preference. This observation leads to applying **rank fusion**, which has been employed for single domain recommenders before (e.g., [2,13,4]), in the context of CDR. By combining the ranked lists from multiple, diverse CDR models, we can integrate their complementary strengths, mitigate their individual weaknesses, and produce a single, unified recommendation list that is more robust and accurate than any of these models alone.

This paper proposes EmbMerge, a new method for this task. Inspired by LambdaMerge [17] but moving beyond its dependence on sparse, hand-crafted features, EmbMerge leverages modern deep learning. It uses four distinct dense vector embeddings to represent the core components of the fusion task: a **Model Vector** for the source CDR model (e.g., BiTGCF [9]), an **Item Vector** for the item being scored, a **Rank Vector** for the item's original position, and a **Score Vector** for the item's normalized score.

These vectors are processed by a simple transformer encoder to learn complex, context-aware interactions between items within the lists generated by different CDR systems. We introduce and evaluate two architectural variants of EmbMerge. Our experiments clearly show that EmbMerge outperforms strong fusion baselines, demonstrating its capability to effectively combine information from multiple cross-domain recommenders.

2 Related Work

The problem of combining multiple ranked lists has been extensively studied in information retrieval and recommendation systems [6,10]. Existing methods can be broadly categorized into unsupervised and supervised approaches.

Unsupervised Methods: These methods use heuristics on ranks or scores without requiring training data. **BordaFuse**, based on the Borda count voting method [1], assigns a score to each item based on its rank (e.g., more points for higher ranks) and sums these scores across all lists. **Reciprocal Rank Fusion (RRF)** [7] calculates a score by summing the *reciprocal* of each item's rank across the lists. These methods are simple and efficient but may not capture complex list interdependencies.

Supervised Methods: These approaches frame fusion as a learning task, using ground-truth data to learn how to best combine lists. **Weighted BordaFuse (W-BordaFuse)** [1] extends the Borda count by learning a specific weight for each input list, thereby assigning more influence to historically more reliable rankers. **Logn-ISR** [11] is another supervised technique that typically uses a regression model.

Our work is inspired by **LambdaMerge** [17] which has been introduced to merge rankings of query reformulations issued to a search engine. LambdaMerge is a supervised method that learns to re-score items based on hand-crafted features describing both the item's position and the list's statistical properties.

While powerful, its reliance on feature engineering limits its ability to model deep semantic patterns. EmbMerge addresses this limitation by replacing these features with learned, contextualized embeddings.

Earlier studies have focused mainly on adapting existing rank fusion techniques to the recommendation domain [2,13,4]. Although these works demonstrate that rank fusion can enhance recommender system performance, our approach differs as we employ dense embeddings to represent lists, items, ranks, and scores, which are then processed by transformer-based architectures in the context of CDR.

3 The Proposed Method: EmbMerge

The main idea of EmbMerge is the use of dense but small, learned embeddings and a transformer encoder architecture to model the relationships between items within and across different recommendation lists. This approach moves beyond the limitations of sparse, statistical features used in methods like LambdaMerge.

3.1 Input Representation

Instead of statistical features, EmbMerge uses four distinct types of dense vectors to represent each item within its source list:

- **Model Vector:** To distinguish which recommendation model produced a given list, each model is assigned a unique, learnable embedding vector.
- **Item Vector:** Each unique item in the dataset is mapped to a learnable embedding vector, allowing the model to capture item-specific characteristics.
- **Rank Vector:** The position of an item in a ranked list is a crucial signal. We use a learnable embedding layer to create a vector for each possible rank (e.g., positions 1 through 100).
- **Score Vector:** Scores from different models can have varying scales and distributions. To handle this, raw scores are first normalized and then passed through a series of projection layers to create a score vector.

For each item in a list, these four vectors are concatenated to form a single input vector.

3.2 Transformer Encoder

The concatenated input vectors for all items in a list are fed as a sequence into a transformer encoder module. The self-attention mechanism of the transformer allows the model to weigh the influence of other items in the list when creating a representation for a specific item. This process results in a sequence of output vectors, where each vector is a context-aware representation of an item, enriched with information about its relationship with other items in the same list.

3.3 Architectural Variants

From the context-aware representations generated by the transformer, we propose two architectures to compute the final scores.

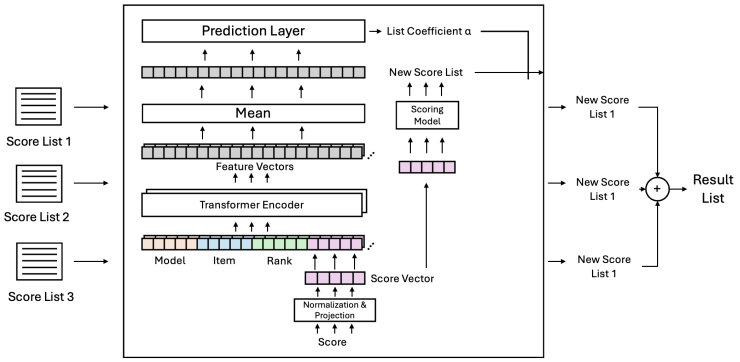

Fig. 1. The architecture of the Standard Model.

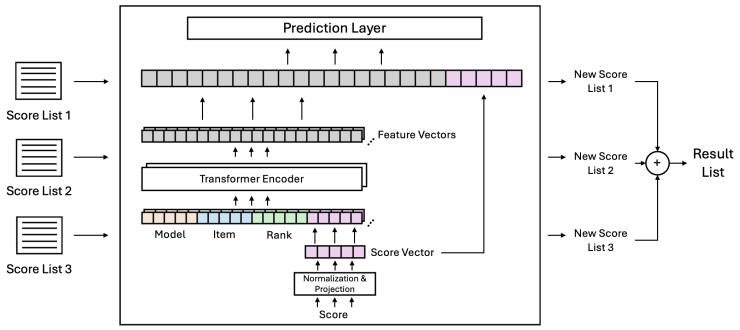

Fig. 2. The architecture of the Direct Prediction Model.

Standard Model. As shown in Fig. 1, this model first computes an aggregated representation of the entire list by taking the mean of all item representation vectors from the transformer encoder. This single list vector is then passed through a prediction layer to generate a list coefficient, α. In parallel, the original score vectors for each item are passed through a small scoring sub-model to generate new base scores. The final score for each item from a given model is the product of its new base score and the list coefficient α. Finally, the newly computed score lists from the three different models are summed up and re-ranked to produce the merged result.

Direct Prediction Model. This architecture, shown in Fig. 2, skips the list-level aggregation step. Instead, the representation vector for *each item* from the transformer encoder is concatenated with its corresponding score vector. This combined vector is then fed directly into a prediction layer to compute the new score for that specific item. This allows the model to make a fine-grained prediction for each item based on its unique context. As with the standard model, the new score lists from the three models are then summed and re-ranked.

For the training of both of these variants, we implemented and used ApproxNDCGLoss [15,5]

4 Experimental Setup

Datasets and Cross-Domain Base Models. The central goal of our experiments is to effectively fuse the outputs of models designed for CDR task. To this end, we conducted experiments on two datasets suitable for CDR: **Digix News Feeds** as source domain & **Digix Ads** as target domain [19] and **Amazon Books** as source domain & **Amazon Movies & TV Shows** [12] as target domain.

To generate the initial ranked lists for fusion, we trained three different cross-domain recommendation systems: **BiTGCF** [9], **CMF** [18] and **DTCDR** [22]. The training of these base models and the generation of score lists were performed using the **Recbole-CDR** [21] library, a specialized framework for reproducing and developing CDR models. To decide which models to use, we trained and tested available models from **Recbole-CDR**, then chose two deeper models with the highest scores (BiTGCF and DTCDR) and a relatively shallow model (CMF). We trained CDR models on each dataset, and generated recommendation lists for each user consisting of 1 positive item and 99 random negative items with the leave-one-out method following the practice in earlier works [9,14,16,20,8]. The generated lists are scored and ranked by three CDR models for the fusion task. In total, 439,636 and 130,649 recommendation lists (per test user/positive item pair) are generated for the Digix and Amazon datasets, respectively.

Baseline Methods. We compared the performance of our EmbMerge models against two unsupervised and two supervised baseline fusion methods: **RRF (Reciprocal Rank Fusion)**, **BordaFuse**, **Logn-ISR** and **W-BordaFuse**. The implementation of these baseline methods was carried out using **ranx.fuse** [3] library with standard parameters assigned by the framework. Four most successful methods implemented by **ranx.fuse** are chosen as base methods. For supervised methods (and also our proposed transformer based approaches); we applied 5-fold cross validation over the recommendation lists generated by the CDR methods for each test user.

Implementation Details. For both EmbMerge models, we used embedding vectors of length 8. The transformer encoder was configured with a single layer, 2 attention heads, an input dimension of 32, and a hidden dimension of 64. The models were trained for 100 epochs on the Digix dataset and 10 epochs on the Amazon dataset. The optimal hyperparameter values were determined over the validation splits. Our implementation is available in our GitHub repository[1].

Evaluation Metrics. We evaluated the performance of all fusion methods using two standard information retrieval metrics: Recall@10 and NDCG@10.

[1] `https://github.com/erdenizayd/EmbMerge`

Table 1. Results on the Digix and Amazon datasets. Base models are cross-domain recommenders. Best fusion performance is in bold.

Method	Digix		Amazon	
	Recall@10	NDCG@10	Recall@10	NDCG@10
Base CDR Models				
BiTGCF	0.8904	0.7129	0.6242	0.3850
CMF	0.8613	0.6729	0.6016	0.3853
DTCDR	0.8780	0.6483	0.5739	0.3423
Fusion Baselines				
W-BordaFuse	0.9038	0.7245	0.6541	0.4134
BordaFuse	0.9011	0.7191	0.6519	0.4095
Logn-ISR	0.8953	0.7086	0.6469	0.4062
RRF	0.9026	0.7187	0.6595	0.4145
EmbMerge Models				
Standard Model	0.9007	0.7237	**0.6619**	**0.4215**
Direct Prediction	**0.9196**	**0.7279**	0.6459	0.4035

5 Results and Discussion

The experimental results for the Digix and Amazon datasets are presented in Table 1. We report the performances of the three individual CDR base models, the baseline fusion methods, and our EmbMerge models.

As shown in Table 1, baseline fusion methods almost always improve upon the performance of the individual cross-domain recommenders. Furthermore, our proposed models outperform all of their competitors for both datasets. On the Digix dataset, **Direct Prediction** model delivers the best overall results, with Recall@10 of 0.9196 and NDCG@10 of 0.7279. Remarkably, our Direct Prediction method yields 1.75% and 0.47% relative improvement in terms of Recall and NDCG metrics, respectively, against the best performing baseline fusion method, i.e., W-BordaFuse. On the Amazon dataset, the **Standard Model** variant achieves a Recall@10 score of 0.6619 and an NDCG@10 score of 0.4215, outperforming the best baseline fusion method (RRF) with relative improvements of 0.36% and 1.69%, in terms of the respective metrics.

Overall, our experiments reveal that across different datasets, either one of our proposed models consistently achieves the best performance and outperforms both supervised and unsupervised fusion baselines. The varying behavior of our models on different datasets may be due to the varying characteristics and sizes of the datasets, which will be investigated in more detail in our future work.

6 Conclusion

In this paper, we introduce EmbMerge, a novel transformer-based supervised method for merging ranked recommendation lists. We specifically demonstrate its effectiveness in fusing the output from multiple cross-domain recommendation systems. By replacing sparse features with learned, small, dense embeddings and

a transformer encoder, EmbMerge successfully models the complex relationships between item ranks and scores. Our experiments on two datasets show that EmbMerge outperforms strong baselines.

As a future work, we plan to explore the application of the EmbMerge framework to other ranking and fusion tasks, both within and beyond the CDR field.

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